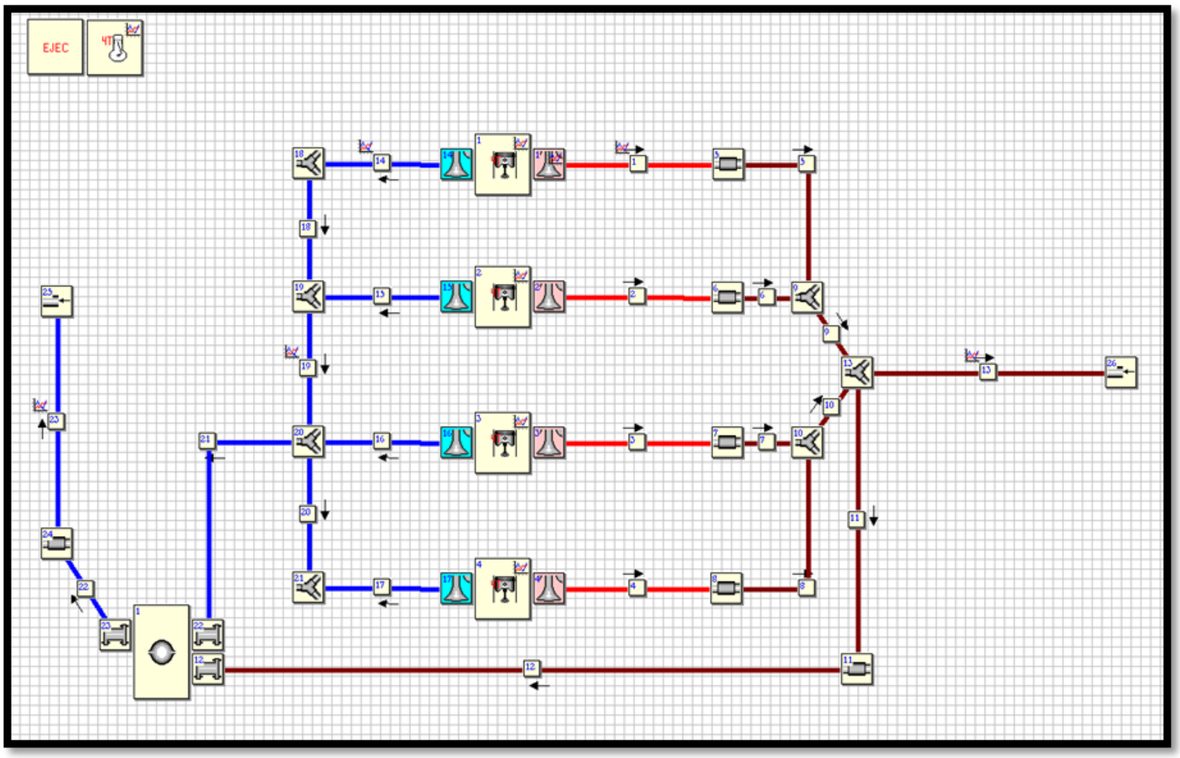
CRITT M2A

Tutorial: Building a 4-Cylinder, 4-Stroke Gasoline Engine Model in OpenWAM 2.0.2

A first approach and introduction to the interface



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Contents

Introduction 3
Setting up the engine 3
Wiebe's law 4
Building the engine: 6
Building the valves: 7
Ducts: 8
Instrumentation 14
Modeling an intercooler: 11
Controlling systems in OpenWAM: 13

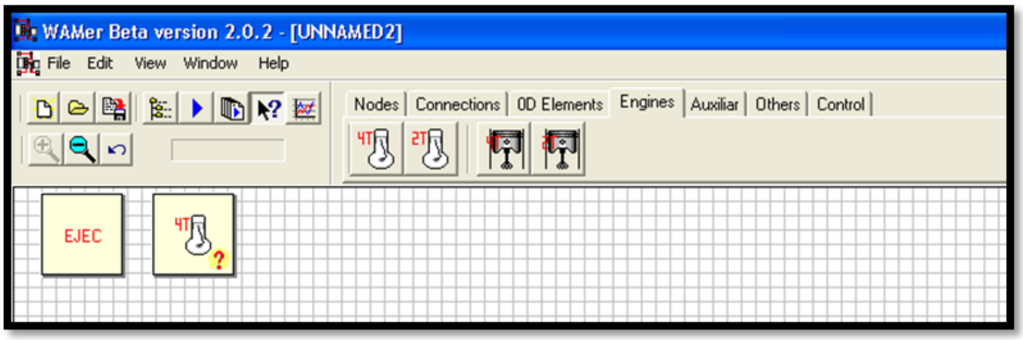
Introduction

This tutorial is designed to familiarize you with OpenWAM and help you develop 1D/0D models for simulation. The case we will present is a naturally aspirated 4-stroke, 4-cylinder, 4-valve engine. The key steps in building the model are:

- Setting up the engine
- Setting up the valves
- Setting up the intake/exhaust ducts
- Finalizing the model, compilation

Setting up the engine

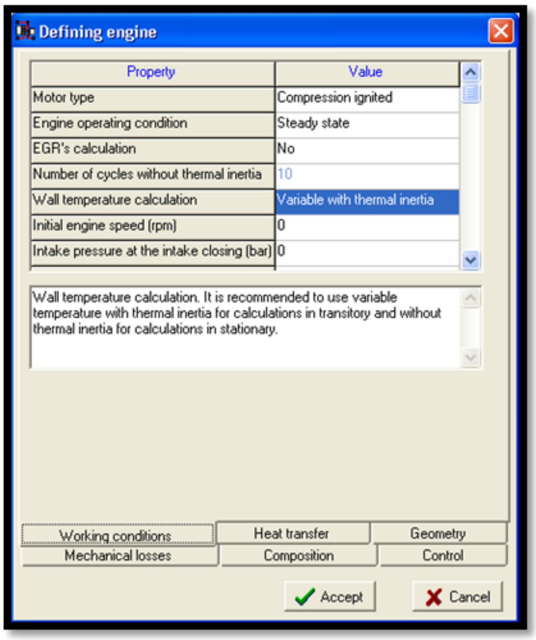
Go to the Engines tab and select the block representing a 4-stroke engine. Drag it into the workspace.



Double-click the engine block and a dialog box will open:

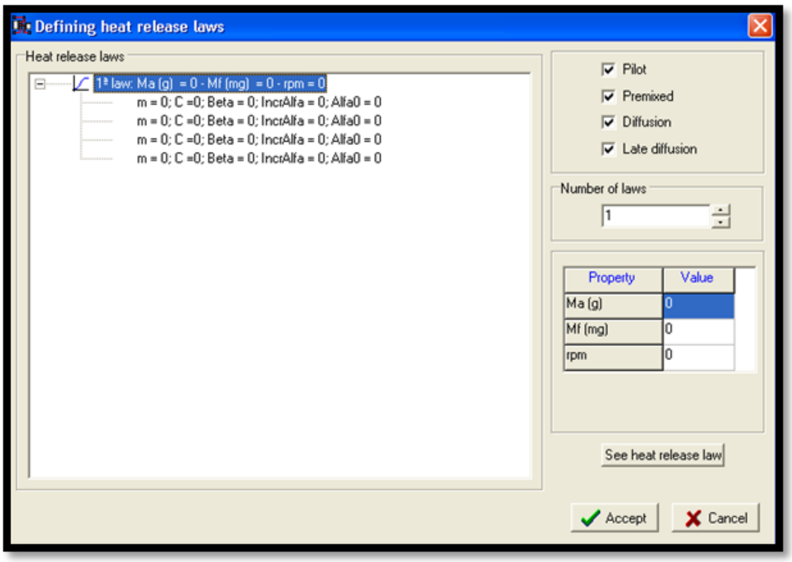
- You can define various settings such as the operating mode, engine rotation speed, mixture characteristics, and engine geometry.

- Left-click the engine block to select: "Define Heat release law"



The following dialog box opens:

It lets you define the coefficients of the heat release law (Wiebe's law).



Wiebe's law

Wiebe's law is defined as follows in OpenWAM:

$$x_b = \beta * \left(1 - \exp\left[-C*\left(\frac{\alpha-\alpha_0}{\Delta\alpha_0}\right)^{m_0+1}\right]\right)$$

Where:

β: represents the proportion of the combustion phase relative to the overall combustion

C: represents the modulation parameter that controls the percentage of fuel injected and burned within the combustion duration

α: the crankshaft angle

α0: the crankshaft angle at which the combustion phase begins

Δα0: the duration of the combustion phase

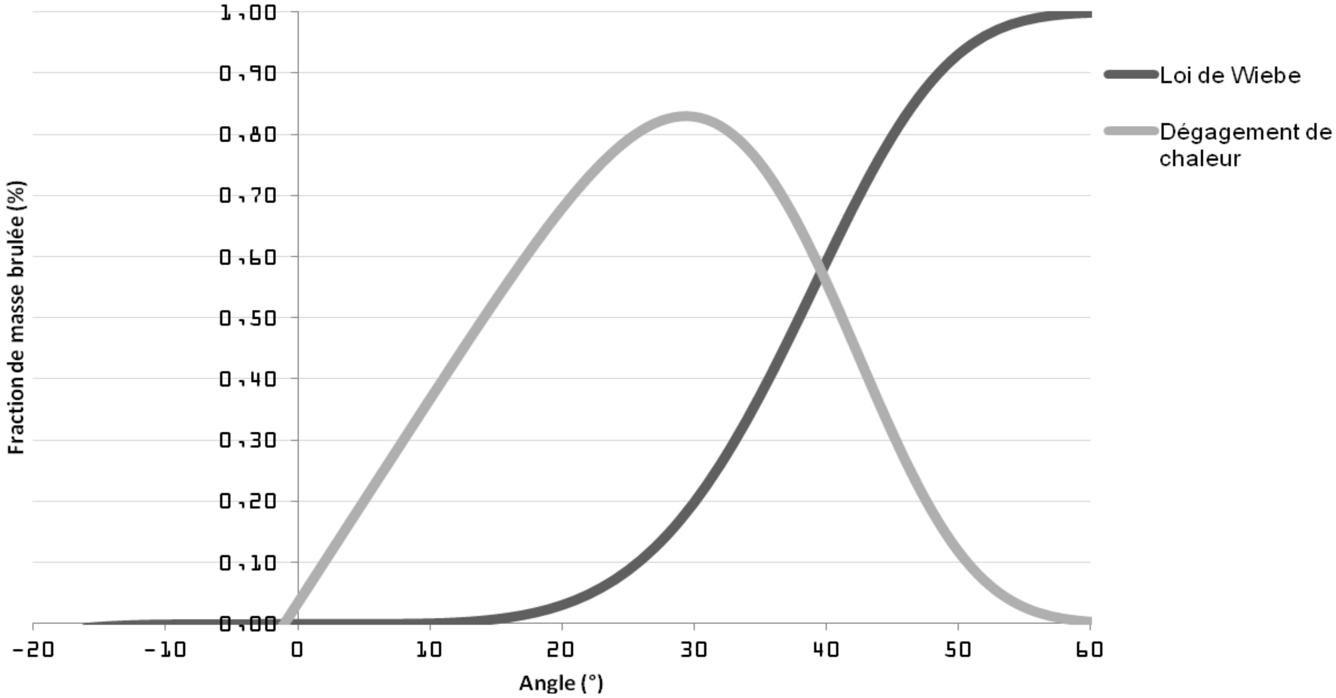
m0: the shape parameter that controls the slope of the combustion phase

combu

When building a Diesel model, integrating the four combustion phases therefore corresponds to the sum of four Wiebe laws.

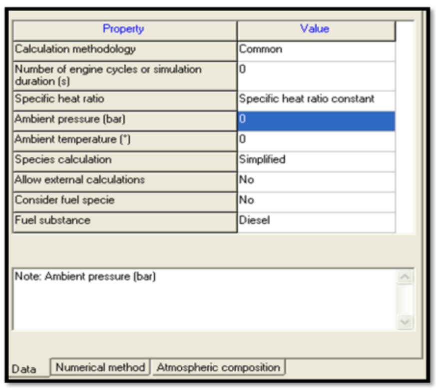
When building a gasoline model, using a single law is possible. The default values that can be used are:

$$0 < \beta < 1;\ C < 6{,}908\ ;\ \alpha_0 = 2\ ;\ \Delta\alpha_0 = 50\ ;\ 0{,}5 < m_0 < 4$$



Model definition block:

To then define the model used and the various calculation parameters, we will select the EJEC block. The data to define are:



- The calculation method: common for a single method, or independent for greater precision based on measured ducts.
- Number of cycles needed to reach convergence
- Pressure and Temperature conditions.
- Calculation type: simplified species (CO_2, C_xH_x, O_2, H_2O, N_2), or extended to include others such as NO, CO, etc.

les CO etc.

- Enabling external calculations, i.e. coupling with MATLAB/Simulink.

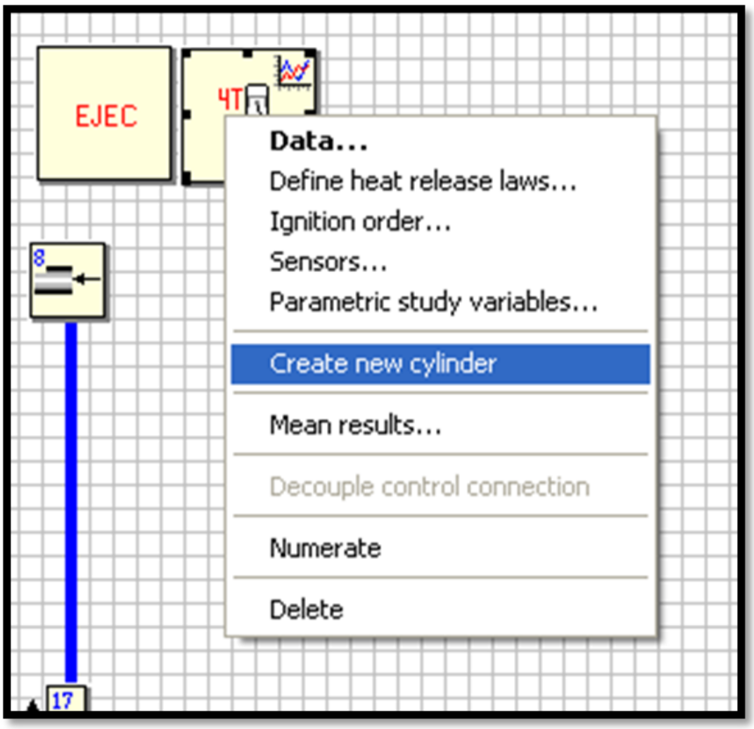
- And finally, the fuel type.

The second tab, Numerical method, lets you select the numerical resolution method used in your model. You can choose between the Lax-Wendroff method, which solves hyperbolic partial differential equations using finite differences; the CE-SE method, a numerical resolution method developed jointly with NASA; and finally the TVD method, for “Total Variation Diminishing”, which provides a second-order solution to hyperbolic partial differential equations without generating oscillations.

In summary, the Lax-Wendroff method produces results more quickly but with less precision for the model. The TVD method is the most precise.

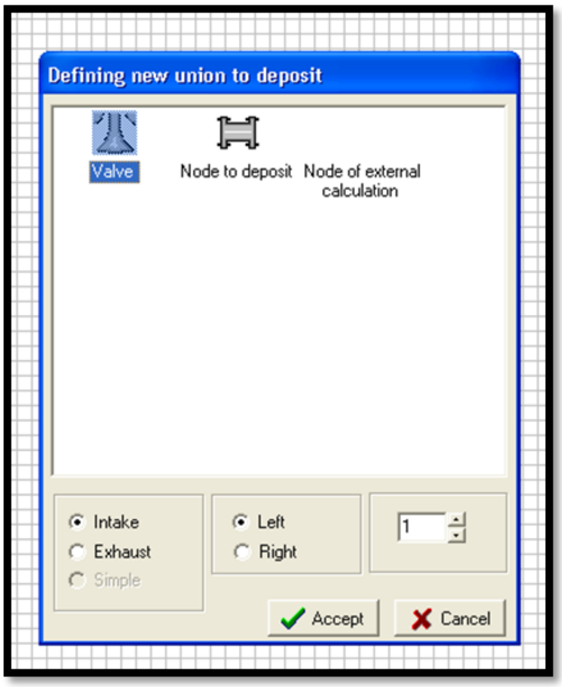
Building the engine:

To build your engine, you need to add cylinders. To do this, right-click the engine block and select “Create new cylinder”. A cylinder block will automatically appear in the workspace.



Next, you'll need to add the valves. Right-click your cylinder: “Create Union”. Select Valve by clicking on it. Then choose your valve type: Intake or Exhaust, and the side on which it will appear.

roit sur votre cylindre : « Create



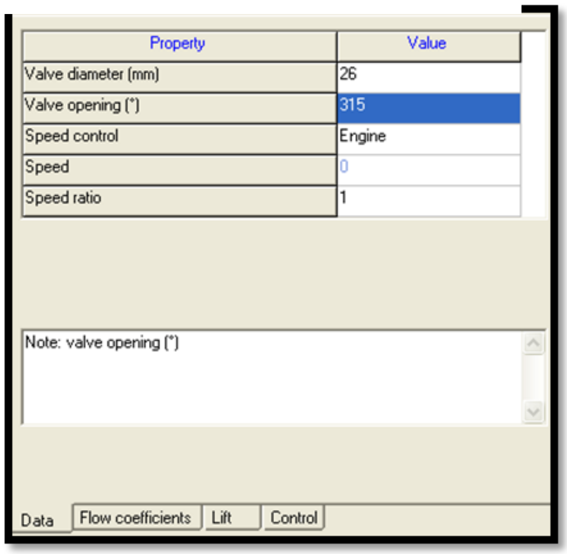
To get a 4-cylinder, 4-valve engine, you'll need to add 4 cylinders and 4 valves per cylinder. You'll then need to add the lift law and the backflow coefficients. Then build your engine architecture by connecting the valves to blocks such as volumes (plenums) and nodes.

Configuring the valves:

Double-click your valve, and the following window appears:

You then need to define the valve diameter, the opening angle from your lift law, and the element controlling the valve speed: here, the engine.

In the flow coefficients tab, you'll need to enter the lift step, the flow diameter, and the flow coefficients.



To enter the flow coefficients and lift laws in the lift tab, we recommend using a text editor (Notepad):

For a lift law, you need to:

- Enter the step between your different values: angle step. OpenWAM will use this to plot the lift law.
- Copy your lift law in mm into a text file.
- Note the total number of values entered at the top of your text file. Example: suppose you have the following lift

bloc note. Exemple : vous avez la loi

law: 0,1,2,3,4,5,6,7,5,3,2,1,0. Your text file will look like this:

13
0
1
2
3
4
5
6
7
5
3
2
1
0

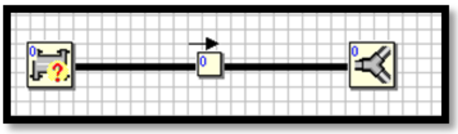
- Note that for decimal values, you must use a period "." as the separator. You can easily do this using the "Edit > Replace All" function in a text editor.

For the flow coefficient table, you must provide the total number of rows.

Each value in each column must be separated by a comma and a space.

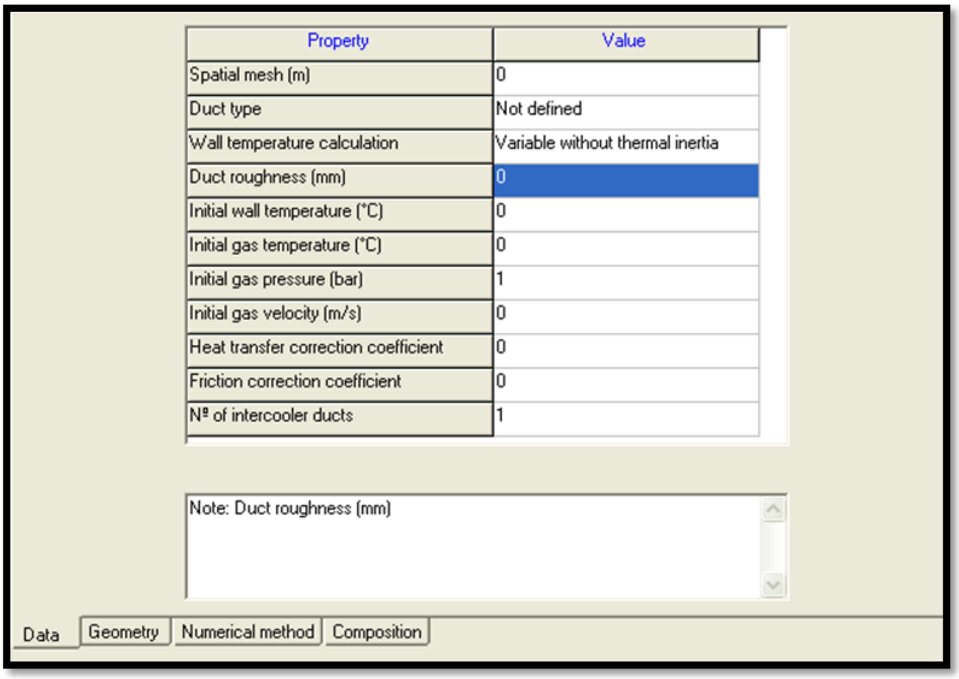
Ducts:

To create a duct, you must first have placed elements on your model that can be connected together, such as a connection or a node.



Click one of your elements, drag it, and release it on the other. A duct symbol will appear and your duct will be created. Clicking on it will bring up the following window:

Spatial mesh: sets the calculation step used in this duct. The smaller it is, the more precise the calculation. However, it will also take longer. Values used for a secondary duct are typically around 1 cm; for more critical ducts, the step should be on the order of a millimeter.



Duct type: here you have four options: intake pipe, intake ports, exhaust pipe, and exhaust ducts. For the intake, we'll use intake pipes; for the exhaust manifold: exhaust ports; and for the exhaust line: exhaust pipes.

Wall Temperature calculation: Here you need to consider whether the duct will act as an intercooler or not. See the explanation provided for modeling an intercooler. Otherwise, choose “constant”.

Duct roughness: defines the roughness of your duct wall. The default value used is 0.15 mm.

Initial Wall Temperature: The wall temperature of your duct. For an intake duct, you should use the T0 temperature you entered in the EJEC block. For a duct located downstream of a compressor, a higher value (around 50°) is recommended. For exhaust ducts, a value close to the exhaust gas temperature is used — approximately 500° to 600° for a diesel engine and 700° to 800° for a gasoline engine.

Initial Gas Temperature: The gas temperature in the duct. Follow the same reasoning as for Initial Wall Temperature.

Initial Gas Pressure: The gas pressure in the duct. Follow the same reasoning as for Initial Wall Temperature.

Initial Gas Velocity: The gas velocity in the duct. A value of 0.01 m/s produces a more stable calculation.

Heat Transfer correction coefficient: the heat transfer correction coefficient.

Friction correction coefficient: the corrected friction coefficient.

N° of intercooler ducts: see the section on intercoolers.

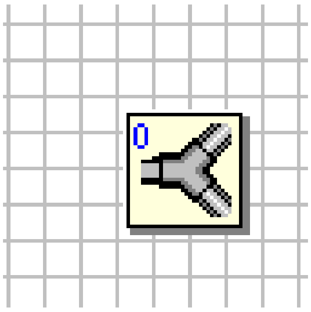
In the geometry tab, you can define your duct's geometry. Inlet diameter corresponds to the entry diameter. By clicking the cross, you can add a length and an outlet diameter to your duct. If you create a duct with different inlet and outlet diameters, you'll get a cone-shaped duct.

Warning

Adding different diameters and lengths to a single duct can cause model instability. We therefore recommend breaking your duct into several sections separated by nodes.

Finally, the Numerical method tab lets you choose the numerical resolution method used, in the same way as in the EJEC block. For an intake duct, the Lax-Wendroff method is sufficient; for an exhaust duct, the TVD method may be more appropriate.

To create a circuit or a manifold, you need the “branch” element in the “nodes” tab. Simply connect your blocks using click-drag-drop.

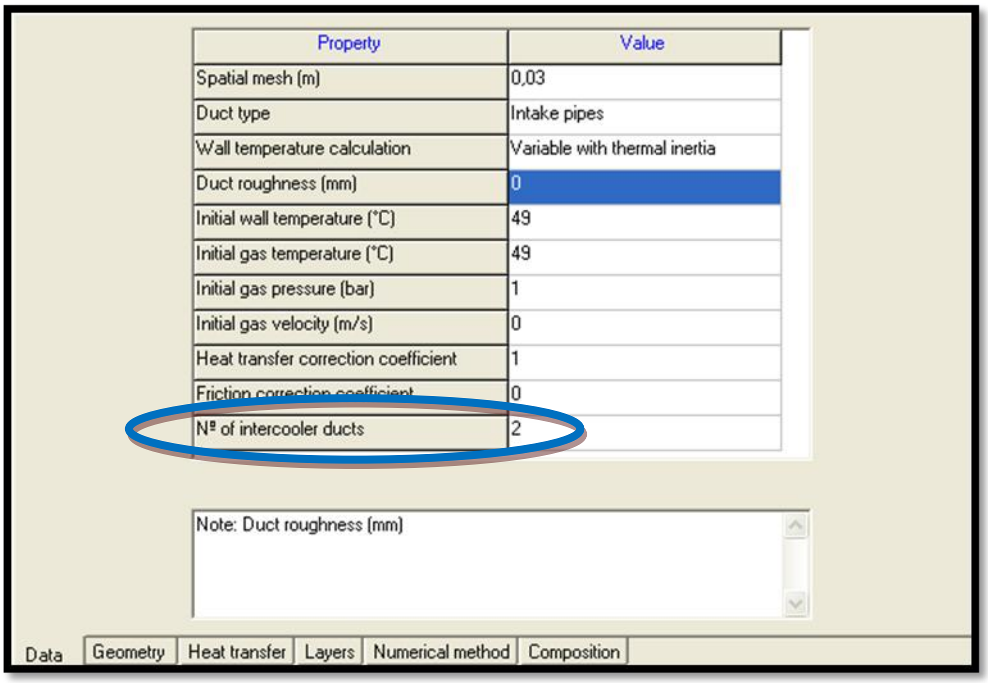


Modeling an intercooler:

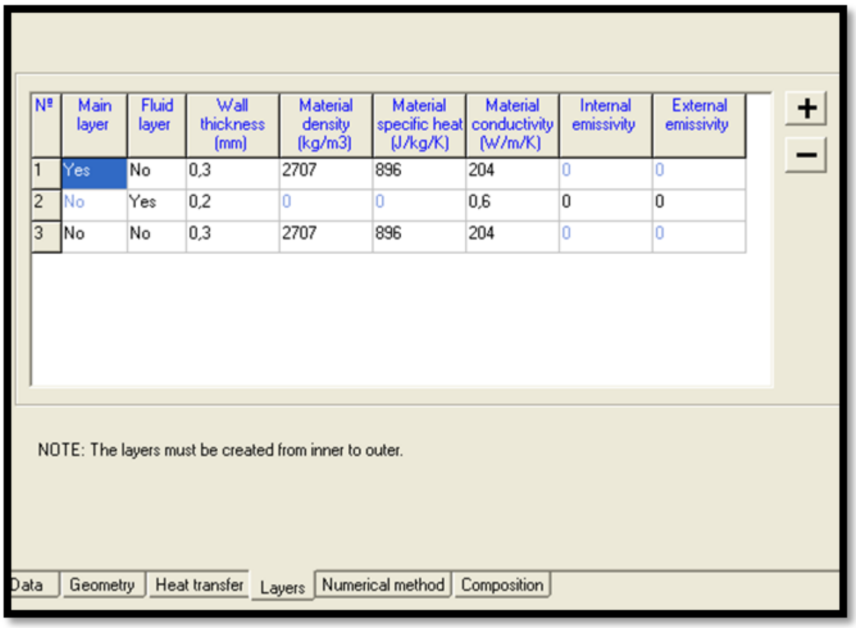
Since OpenWAM doesn't offer a heat exchanger block, you'll need to model an intercooler as follows:

To start, you need to model a duct between two elements. See the duct modeling section for more details. This duct will become a heat exchanger:

You then need to change the value for the number of ducts.

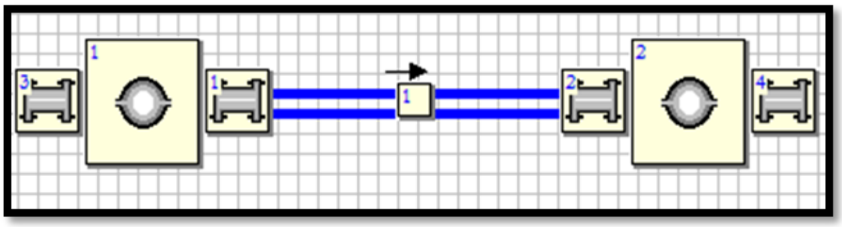


For better results, change the Wall temperature calculation setting and adjust the duct values across different layers. An example is shown below to illustrate this.



Nº	Main layer	Fluid layer	Wall thickness (mm)	Material density (kg/m3)	Material specific heat (J/kg/K)	Material conductivity (W/m/K)	Internal emissivity	External emissivity
1	Yes	No	0,3	2707	896	204	0	0
2	No	Yes	0,2	0	0	0,6	0	0
3	No	No	0,3	2707	896	204	0	0

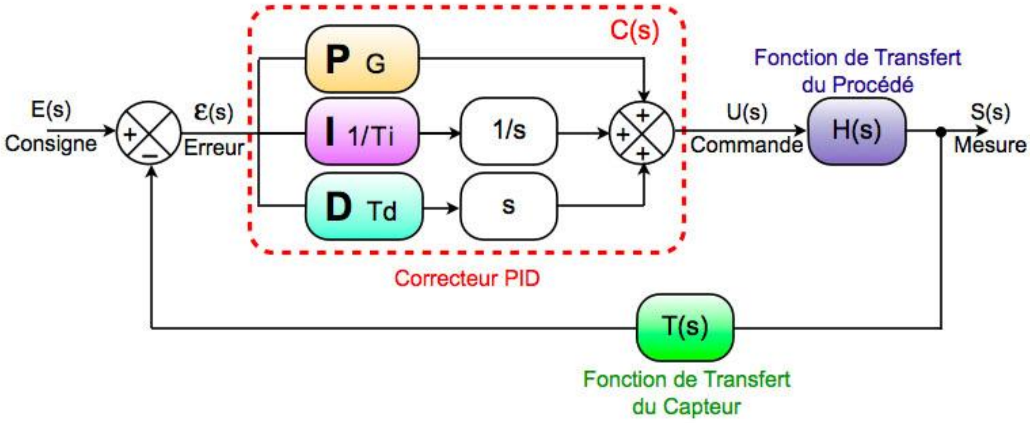
And here is an example of an intercooler architecture you could build.



To achieve the desired result, we recommend adjusting the “initial Wall” and “initial gas” temperatures, as well as the passage lengths and diameters.

Controlling systems in OpenWAM:

Here we'll show you how to implement a PID control block:

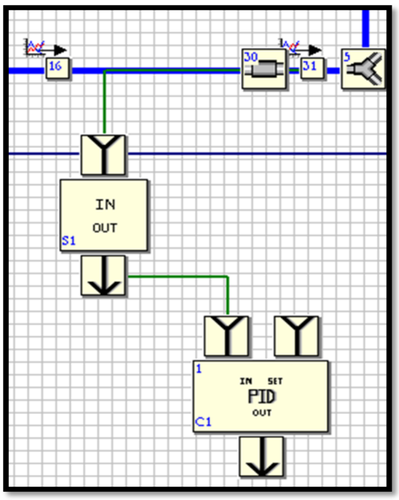


To start, go to the control tab in the OpenWAM workspace.

You'll need to add two elements: the In/Out block and the PID block.

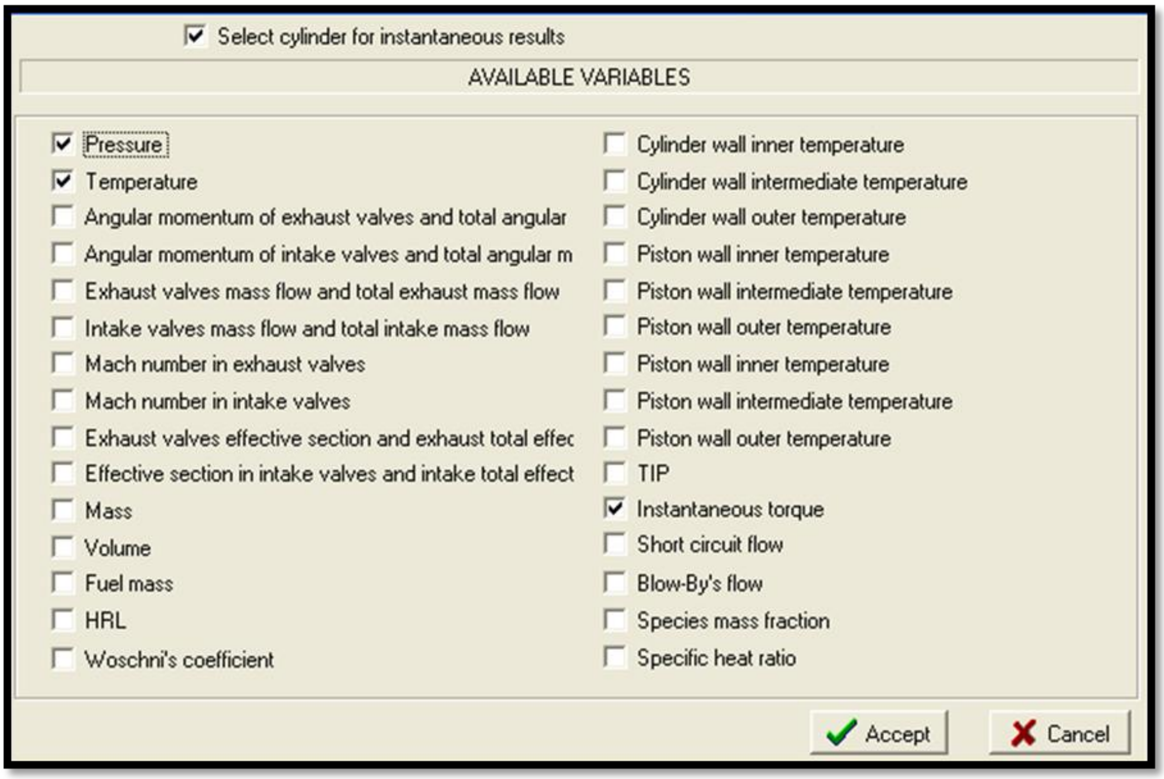
The In/Out block must be connected to a sensor. To do this, go to the element you want to measure. Right-click, sensors. You can select the physical quantity to measure, and set a gain and a measurement time. Connect the In/Out block's input to your sensor. Then connect the block's output to the PID block's “in” input. You can then configure your PID block.

Finally, connect the PID output to the controlled element, a “throttle valve” for example. A small red C will appear, showing that your element is under the control of the PID block.



Instrumentation

Now that your engine architecture is complete, you can instrument it to run your simulations. To do this, select the element you're interested in. Right-click and choose “mean results” or “instantaneous results”. You'll get a window similar to this:



Check “select [element] for instantaneous results” and you can then choose the variables you want to measure.

Once you've finished instrumenting, there's one final step required for your simulation: go to Edit and select “Numerate”. OpenWAM will then assign a number to each element present. If you prefer, you can do this step manually by right-clicking your block and selecting numerate manually.

Your model is now ready to be compiled.

Final example

Here is an example of a turbocharged 4-cylinder, 4-valve engine, controlled with a PID, shown in OpenWAM: